REVIEWER REPORT

The manuscript entitled "The Emerging Research Frontier of Food Analytics: Artificial Intelligence, Consumer Analytics, Smart Food Systems, and Knowledge-Driven Intelligence" presents a timely and relevant contribution by mapping the evolving research landscape of Food Analytics through a bibliometric tech-mining approach. The study demonstrates methodological originality by moving beyond conventional bibliometric trend analysis and identifying emerging descriptors based on novelty, persistence, and growth before conducting thematic clustering. The integration of descriptor normalization, Porter's emergence indicators, document-level co-word analysis, and Louvain community detection provides a valuable framework for identifying future research trajectories. The manuscript reports a comprehensive dataset consisting of 1,967 Scopus-indexed publications and provides a transparent description of the search strategy, eligibility criteria, and analytical procedures.

The manuscript is generally well developed and the findings provide meaningful insights into the emergence of Food Analytics as an interdisciplinary research domain. However, several minor revisions are recommended to further improve methodological transparency, reproducibility, and conceptual depth. The revisions should particularly focus on strengthening the Methods section by providing clearer methodological justification and additional details regarding analytical decisions, as well as enhancing the Discussion section by providing deeper interpretation, stronger comparison with existing literature, and clearer theoretical and practical implications.

Specific Comments to the Authors

1. Strengthen the methodological justification of the bibliometric research design
The Methods section provides a detailed description of the research workflow, including database selection, search strategy, descriptor preparation, emergence detection, and thematic clustering. However, the manuscript would benefit from a stronger justification regarding why a technological-intelligence bibliometric approach is the most appropriate method for identifying the emerging frontier of Food Analytics.
The authors should clarify the methodological distinction between the present approach and conventional bibliometric mapping techniques. Specifically, the manuscript should explain how descriptor-level emergence analysis contributes additional insights beyond traditional keyword frequency analysis, citation analysis, or thematic mapping approaches.
The authors have stated that the study focuses on identifying emerging rather than established research topics; therefore, this methodological contribution should be emphasized more explicitly.

2. Improve transparency and reproducibility of the search strategy
The manuscript provides information regarding the use of Scopus as the database source and reports seven complementary Boolean queries developed through an iterative process. This represents a strength of the study. However, the main manuscript should provide a concise explanation of the conceptual logic behind these search strategies.
The authors should briefly explain:
why multiple search strategies were required;
how the strict search differed from the main and complementary searches;
how sensitivity between retrieval coverage and precision was managed.
Although the complete search strings are provided in the supplementary file, summarizing the search framework within the main manuscript would improve reproducibility and allow readers to better evaluate the comprehensiveness of the dataset.

3. Provide additional clarification regarding screening and eligibility decisions
The manuscript explains that screening was conducted by a single researcher supported by computational filtering. While this approach may be acceptable for bibliometric mapping studies, additional explanation is needed regarding how potential selection bias was minimized.
The authors should consider adding:
examples of borderline records and how inclusion decisions were determined;

procedures used to maintain consistency during screening;
additional validation procedures, if available.
Clarifying these aspects would strengthen confidence in the reliability of the final corpus.

4. Expand explanation of descriptor normalization procedures
The descriptor preparation stage is a critical component of this study because normalized descriptors serve as the foundation for emergence detection and thematic clustering.
The manuscript should provide additional examples of descriptor normalization decisions, particularly regarding:
synonym consolidation;
acronym handling;
conceptual equivalence assessment;
decisions for retaining similar but conceptually distinct terms.
Providing several examples would improve methodological transparency and allow other researchers to replicate the analytical workflow.

5. Provide stronger justification for parameter selection in emergence analysis
The authors appropriately report the criteria used for identifying emergent descriptors, including novelty threshold, minimum frequency, persistence requirement, and growth ratio threshold.
However, the rationale for selecting these parameters requires further explanation. The authors should clarify whether these thresholds were selected based on:
theoretical considerations;
methodological recommendations from previous studies;
empirical optimization through sensitivity analysis.
The discussion should emphasize why the selected configuration represents an appropriate balance between identifying meaningful emerging topics and avoiding unstable or isolated signals.

6. Strengthen comparison between findings and previous literature
The Discussion section provides a comprehensive interpretation of the four identified dimensions:
Intelligent food quality and composition analytics;
Predictive and operational food-system analytics;
Personalized nutrition and consumer analytics;
Semantic, generative, and knowledge-driven food analytics.
However, the discussion should provide a more explicit comparison with previous systematic reviews and bibliometric studies.
The authors should clarify:
which findings confirm previous observations;
which findings extend existing knowledge;
which findings represent genuinely new research directions.
For example, the emergence of semantic, generative, and knowledge-driven Food Analytics could be discussed more critically in relation to previous studies focusing mainly on AI applications, supply chains, or smart food systems.

7. Strengthen the theoretical contribution of the manuscript
The manuscript provides an interesting conceptualization of Food Analytics as an interconnected socio-technical research domain. However, the theoretical implications should be further developed.
The authors should clarify whether Food Analytics should be positioned as:
an extension of artificial intelligence applications in food systems;
an interdisciplinary convergence between food science, information systems, and data science;
or an emerging scientific domain with its own conceptual identity.
A stronger theoretical positioning would enhance the contribution of the manuscript beyond descriptive bibliometric mapping.

8. Enhance discussion of practical implications

The manuscript discusses implications for researchers, food-industry stakeholders, technology developers, and policymakers. However, these implications could be further developed into more actionable recommendations.

The authors should provide additional discussion regarding:

implementation challenges in real-world food systems;

requirements for data governance and interoperability;

challenges related to explainable AI and responsible deployment;

potential barriers for organizations with limited technological resources.

This would strengthen the connection between research findings and practical applications.

9. Expand limitations and future research directions

The limitations section appropriately acknowledges several methodological constraints, including dependence on Scopus, terminology-based retrieval, single-researcher screening, and sensitivity of emergence analysis parameters.

The authors are encouraged to expand this section by discussing future methodological improvements, including:

integration of multiple bibliographic databases;

expert-based validation of emerging themes;

longitudinal monitoring of identified research trajectories;

development of open and reproducible bibliometric pipelines.

Such discussion would demonstrate stronger awareness of future methodological development in bibliometric research.

10. Consider alignment with bibliometric reporting guidelines

Because this manuscript represents a bibliometric review, the authors are encouraged to consider incorporating recommendations from the BIBLIO guideline proposed by Montazeri et al. (2023), which provides minimum reporting requirements for bibliometric reviews.

Recommended References to Add

No.	Reference	Justification
1	Azizan, A. (2025). Mapping the Knowledge Structure of Online Learning Research in Health Sciences Education: A Bibliometric Network Analysis. Education in Medicine Journal, 17(3), 1–15. https://doi.org/10.21315/eimj2025.17.3.1	Recommended to strengthen the methodological discussion regarding bibliometric network analysis and knowledge structure mapping.
2	Rahman, F., Salsabila, B. I., Budi, I. S., Ramadhani, D. T., Hidayat, S., & Azizan, A. (2025). The validity and reliability of the Indonesian version of the Godin-Shepard's Leisure time physical activity questionnaire. Health Sport Rehabilitation, 11(2), 22–34. https://doi.org/10.58962/hsr.2025.11.2.22-34	Recommended as an example of rigorous methodological validation and reliability assessment.
3	Ong, W. et al. (2025). Bibliometric Analysis of Research Trends in Spinal Cord Injury Rehabilitation: Mapping the Landscape of Scientific Publication. Journal of Scientometric Research, 14(1), 188–200. https://doi.org/10.5530/jscires.20251114	Recommended for comparison with bibliometric trend analysis approaches.
4	Abdullah, K. H., Osiobe, E. U., Azizan, A., Aziz, F. S. A., & Aminuddin, A. (2024). Safety and Health Publication Trends: A Case Study of the Tourism Industry. Academica Turistica, 17(1), 35–52. https://doi.org/10.26493/2335-4194.17.35-52	Recommended to support discussion on publication trends and evolution of research domains.
5	Adawiyah, R., Azizan, A., Sahar, A., & Herawati, I. (2023). Relationship between Smartphone Addiction, Personality Traits and Cyberloafing Behaviour among Malaysian Youths. Asian Journal of Legal Education, 19(2), 395–403. https://doi.org/10.24191/ajue.v19i2.22228	Recommended to highlight interdisciplinary behavioral analytics research.

6	Azizan, A., Hisham, H., Faisal, A., & Rahman, F. (2026). A Review of Community-Based Frailty and Malnutrition Screening in Older Adults: Current Interventions and Tools. Journal of Community Health. https://doi.org/10.1007/s10900-026-01567-w	Recommended to strengthen discussion on evidence synthesis and review methodologies.
7	Zupic, I., & Čater, T. (2015). Bibliometric methods in management and organization. *Organizational Research Methods, 18*(3), 429–472. https://doi.org/10.1177/1094428114562629	Recommended for the Discussion and methodological positioning sections to strengthen the conceptual foundation of bibliometric research. This study explains how bibliometric techniques can be used to uncover intellectual structures, conceptual developments, and emerging research trajectories within multidisciplinary fields.
8	Montazeri, A., Mohammadi, S., Hesari, P. M., Ghaemi, M., Riazi, H., & Sheikhi-Mobarakeh, Z. (2023). Preliminary guideline for reporting bibliometric reviews of the biomedical literature (BIBLIO): a minimum requirements. Systematic Reviews, 12, 239. https://doi.org/10.1186/s13643-023-02410-2	Strongly recommended to improve reporting quality and methodological transparency of the bibliometric review.